Structural constraints on reductions of elliptic witnesses

Abstract

We examine algebraic reductions of a rational-source construction whose integer labels are carried by multiples of a point over a quadratic number field. We prove dimension bounds for recovering arbitrary tuples of such points, the Zariski density of the graph of the integer label, and an obstruction to replacing the paired conjugate reductions by one rational elliptic curve. A related two-curve obstruction applies when a curve correspondence is used to enforce a common integer index. We also identify the exact squareclass restriction on a shared pole parameter and give an explicit counterexample to a parity shortcut in scalar height arguments. These statements concern the specified strategies; they are not lower bounds for arbitrary Diophantine definitions.

1 Point labels and finite algebraic presentations

Let $K = \mathbb{Q}(\sqrt{2})$ and

$$D = 11 - 6\sqrt{2}, \qquad E : y^2 = x^3 - Dx,$$
$$P = \left(\frac{57 - 28\sqrt{2}}{8}, \frac{-660 + 417\sqrt{2}}{32} \right).$$

Let $A = \operatorname{Res}_{K/\mathbb{Q}} E$ be the Weil restriction, and identify $E(K)$ with $A(\mathbb{Q})$. The elliptic-chart argument in [1] proves that A is simple over $\mathbb{Q}$; in particular its proof distinguishes the two conjugate curves by their good reductions at 17. The point P has infinite order. Write $\Gamma = \mathbb{Z}P \subset A(\mathbb{Q})$.

Lemma 1.1. *The subgroup Γ is Zariski dense in A, and Γ^r is Zariski dense in A^r for each positive integer r.*

Proof. The Zariski closure of a subgroup of an algebraic group is an algebraic subgroup. Its identity component is defined over $\mathbb{Q}$. Since P has infinite order, this component has positive dimension; simplicity forces it to be A. Density of the Cartesian powers follows successively by restricting a closed subset to fibers in one factor. □

Theorem 1.2. *Suppose finitely many algebraic varieties Y_j over $\mathbb{Q}$ admit finite morphisms to locally closed subvarieties of $\mathbb{A}^b$, and morphisms $F_j : Y_j \to A^r$. If*

$$\Gamma^r \subset \bigcup_j F_j(Y_j(\mathbb{Q})),$$

then $b \geq 2r$.

Proof. Each Y_j has dimension at most b. The finite union of the closures of the images of the F_j contains Γ^r, and therefore equals A^r by Lemma 1.1. Since A^r is irreducible, one of these image closures is all of A^r. Consequently $2r = \dim A^r \leq \dim Y_j \leq b$. □

The theorem applies to algebraic presentations in which b rational coordinates are retained and all further coordinates are finite algebraic choices. Localizing by a nonzero guard or taking finitely many charts does not change the argument. In particular, recovering arbitrary tuples of nine point labels in this way requires at least eighteen retained rational parameters. If an input coordinate and one final finite-algebra coordinate are counted separately, fewer than ten total coordinates permit at most three independent point labels in this generic recovery model.

This conclusion is deliberately limited. Completeness of a Diophantine construction need not require arbitrary tuples of labels, and its arithmetic witnesses can satisfy useful dependencies. The theorem does not exclude such dependencies or a different integer-label mechanism.

2 The integer index is not an algebraic coordinate

Theorem 2.1. *Let B be a positive-dimensional abelian variety over a field of characteristic zero, and let $Q \in B$ generate a Zariski-dense cyclic subgroup. Then*

$$\{(n, [n]Q) : n \in \mathbb{Z}\}$$

is Zariski dense in $\mathbb{G}_a \times B$. The same assertion holds with n restricted to any nonconstant arithmetic progression.

Proof. The displayed set is the cyclic subgroup generated by $(1, Q)$. Let G be its Zariski closure, and pass to the identity component G^0. Both projections of G^0 are surjective: the images of G are dense subgroups of the respective factors, and the finite component group cannot account for a positive-dimensional connected quotient.

The kernel of $G^0 \to B$ is an algebraic subgroup of $\mathbb{G}_a$, hence is either trivial or all of $\mathbb{G}_a$ in characteristic zero. In the first case the surjective homomorphism has finite kernel and is an isogeny of connected algebraic groups. Such a homomorphism is finite, so G^0 is proper because B is proper. Every morphism from a connected proper variety to the affine line is constant, contradicting the surjective projection to $\mathbb{G}_a$. Thus the kernel contains $\mathbb{G}_a$. Surjectivity to B then gives $G^0 = \mathbb{G}_a \times B$.

For an arithmetic progression $n = a + dm$, with $d \neq 0$, the closure of $\mathbb{Z}[d]Q$ is again all of B: a proper closed subgroup containing it would have finitely many translates containing $\mathbb{Z}Q$. The preceding argument applied to $(d, [d]Q)$ proves the assertion, after translation by $(a, [a]Q)$. □

Corollary 2.2. *On a nonempty affine open subset of A, no nonzero polynomial relation with coefficients in its coordinate ring holds between the integer n and the point $[n]P$ for all integers n for which the point belongs to that open set. In particular, an algebraic finite extension of the point coordinates cannot recover every integer label.*

Proof. Any such relation would cut out a proper closed subset of the product of the open set with $\mathbb{A}^1$, contrary to Theorem 2.1. □

For example, the Lucas relation

$$t^2 - a(a-2)tb + (a-2)^2b^2 - 1 = 0$$

is monic in t. If a, b are already integers and t is rational, the rational-root theorem gives $t \in \mathbb{Z}$. In the elliptic source, however, a, b, t are integer indices of points, rather than available rational coordinates. Replacing a point witness by this monic argument requires a separate way to make the relevant indices available. Corollary 2.2 rules out doing so by a uniform finite algebraic recovery of all labels.

3 The paired-place mechanism over a rational curve

Let $E_0/\mathbb{Q}$ be an elliptic curve, let $P_0 \in E_0(\mathbb{Q})$ be non-torsion, and let $K/\mathbb{Q}$ be quadratic. At a rational prime p of good reduction which splits in K, the two residue fields are canonically $\mathbb{F}_p$. A point belongs to the formal group at one of the two places exactly when its reduction there is the identity.

Proposition 3.1. *If $H = [h]P_0$ is formal at one place above p, and $H - P_0$ is formal at the other, then P_0 reduces to the identity modulo p. Only finitely many good rational primes can satisfy this conclusion for a fixed affine point P_0.*

Proof. The reductions of P_0 at the two places are the same rational point $\overline{P}_0 \in E_0(\mathbb{F}_p)$. The two requirements give

$$[h]\overline{P}_0 = O, \qquad [h-1]\overline{P}_0 = O.$$

Subtracting yields $\overline{P}_0 = O$. For a fixed affine point this occurs only at primes dividing the denominators of its coordinates in a fixed integral model, together with the finitely many excluded primes. □

Thus a rational elliptic curve cannot replace the quadratic-field curve while retaining this paired-place test at arbitrarily many contact primes. The proposition concerns this mechanism, not arbitrary constructions on rational elliptic curves.

3.1 Two rational curves with a common index

Proposition 3.2. *Let E_1, E_2 be elliptic curves over $\mathbb{Q}$ and let $P_i \in E_i(\mathbb{Q})$ be non-torsion. If an algebraic curve in $E_1 \times E_2$ contains*

$$\{([n]P_1, [n]P_2) : n \in \mathbb{Z}\},$$

then E_1, E_2 are isogenous over $\mathbb{Q}$. There are an isogeny $\phi : E_1 \to E_2$, a nonzero integer a, and a torsion point T such that

$$\phi(P_1) = aP_2 + T. \tag{1}$$

For such a relation, coprimality of the two reduction orders can occur at only finitely many good rational primes.

Proof. The Zariski closure of the displayed cyclic subgroup is an algebraic subgroup of dimension at most one. Its identity component G has dimension one because each P_i is non-torsion. The two projections $G \to E_i$ are isogenies. Let the first projection have degree $a > 0$, and compose its dual isogeny with the second projection to obtain $\phi : E_1 \to E_2$. Some nonzero integer multiple of (P_1, P_2) lies in G. The defining property of the dual isogeny then shows that $\phi(P_1) - aP_2$ is torsion, proving (1).

Let m be the order of T, let $d = \deg \phi$, and let k_i be the reduction order of P_i at a good prime where the isogeny and the relation reduce. Multiplying the relation gives

$$k_2 \mid amk_1.$$

Also $[mk_2]P_1$ lies in the kernel of the reduced isogeny, which is killed by d, and hence

$$k_1 \mid dmk_2.$$

If $\gcd(k_1, k_2) = 1$, these imply $k_1 \mid dm$ and $k_2 \mid am$. Thus both orders are bounded independently of the prime. For a fixed non-torsion point, bounded reduction order occurs only at finitely many primes: each such prime makes one of finitely many fixed nonzero multiples reduce to the identity. □

In particular, two nonisogenous rational curves may have the needed distinct reduction orders, but their common-index pairs cannot be recovered from a curve correspondence. When a curve correspondence does exist, the isogeny relation prevents arbitrarily many coprime-order reductions. A two-curve method would therefore need a different way to enforce or replace the common integer index.

4 Sharing a pole parameter

The first equation of each pole certificate is

$$p_i^2 = B_i(1 - e_i)(1 + 3e_i), \qquad B_i \neq 0.$$

Proposition 4.1. *If two such equations share $e_i = e$ and $p_1 p_2 \neq 0$, then B_1/B_2 is a rational square. If one of p_1, p_2 is zero, then both are zero and $e \in \{1, -1/3\}$.*

Proof. In the nonzero case division gives $B_1/B_2 = (p_1/p_2)^2$. Otherwise $(1 - e)(1 + 3e) = 0$, giving exactly the stated alternatives. □

The single rational chart used in the source is

$$e_i = \frac{\tau_i^2 - B_i}{\tau_i^2 + 3B_i}, \qquad p_i = -\frac{4B_i\tau_i}{\tau_i^2 + 3B_i}.$$

Sharing τ_i is a different proposal: it need not share e_i, so Proposition 4.1 does not apply. It does, however, impose a simultaneous rational-point requirement on the remaining elliptic quartics. The inherited realization theorem chooses a fresh twist parameter for each prepared pole input. It does not assert that an arbitrary fixed or shared chart parameter is available.

The degenerate value $e = -1/3$ is allowed by this chart, with $\tau = 0$. Consequently the squareclass obstruction alone does not exclude fixing this value. Completeness of that specialization would require a new rational-point theorem for its remaining quartic; no such completeness conclusion is made here.

5 Odd labels can be formal at both conjugate places

Proposition 5.1. *For the curve and point of Section 1, both conjugate reductions of P at the split good prime $p = 77017$ have odd order. More precisely,*

$\sqrt{2}$	D	P	$\mathrm{ord}(P)$
18810	41185	$(1562, 38114)$	2141
58207	35854	$(56215, 58116)$	2405.

The odd integer $5149105 = 2141 \cdot 2405$ therefore places its multiple of P in both conjugate formal groups simultaneously. Furthermore

$$p > 76176, \qquad p \equiv 1 \pmod 8, \qquad p \equiv 3 \pmod 7.$$

Proof. Trial division through $\lfloor\sqrt{p}\rfloor$ proves that p is prime. Direct modular arithmetic verifies that the two displayed images of $\sqrt{2}$ square to 2, and gives the indicated curve coefficients and point coordinates. Both curves have good reduction. Write P_1, P_2 for the two reduced points. Exact elliptic-curve addition gives

$$[2141]P_1 = O, \qquad [2405]P_2 = O.$$

Since 2141 is prime and $P_1 \neq O$, the first order is exact. For the second, $2405 = 5 \cdot 13 \cdot 37$, and the proper prime-quotient multiples are

$$[481]P_2 = (11050, 75952),$$
$$[185]P_2 = (14926, 42895),$$
$$[65]P_2 = (74297, 22537).$$

Each is nonzero, proving that the second order is exact. The two orders are coprime and odd, so their product kills both reductions. Finally the three stated comparisons and congruences are immediate integer calculations.

The accompanying standard-library verifier

`check_large_split_orders.py`

reproduces every calculation by integer modular arithmetic. As an independent consistency check, it counts the two curves both by a square-root frequency table and by Euler's criterion, obtaining cardinalities 77076 and 76960, respectively. The resulting exact witnesses are recorded in

`large-split-order-certificate.json`.

□

Thus the shortcut that parity of an integer label singles out one formal embedding fails even above the exceptional cutoff, for a prime satisfying the two displayed rational congruences. These congruences alone do not assert the stronger Gaussian-generator or avoidance conditions of the simultaneous prime-pattern theorem. The example makes no claim about those additional conditions and is not a counterexample to any proved aggregate height comparison.

The same verifier also gives odd orders 2377 and 19109 at the fixed scalar prime $\ell = 76249$. A separate complete enumeration below 3000 finds 209 split good primes, with odd orders at both places for 72 of them; its exact computation is supplied in

`search_reduction_parities.py`.

6 Rational injections and finite fibers

Pasten constructs a regular injection on rational points from a suitable affine surface $V \times V$, with V an open part of a positive-rank elliptic curve, to the affine line [3]. Such a rational-point injection need not have finite geometric fibers. Indeed any dominant morphism from a surface to a curve has positive-dimensional generic fiber. It therefore does not by itself provide a finite-algebra presentation with a single retained coordinate.

Likewise, the polynomial injections of two rational variables studied by Poonen [4] concern injectivity on rational points. The stated existence theorem assumes a rational-point nondensity hypothesis. Neither injectivity nor this additional hypothesis supplies the finite geometric fibers needed by the finite-root elimination theorem. An application to witness compression must provide that separate step.

References

[1] *An obstruction to one-parameter presentations of an elliptic point*, project paper, version 7, 7 November 2026.

[2] *A defining polynomial in twenty-seven variables: an existence theorem with a uniform degree bound*, project paper, version 12, 9 November 2026.

[3] H. Pasten, *Bivariate polynomial injections and elliptic curves*, arXiv:1811.02416, `https://arxiv.org/abs/1811.02416`.

[4] B. Poonen, *Multivariable polynomial injections on rational numbers*, Acta Arith. **145** (2010), 123–127, `https://math.mit.edu/~poonen/papers/QQ.pdf`.